Eduardo Figueroa Medina
Oct. 14, 2026
Data Science Capstone

Section Draft - Introduction

As perhaps the most important river in the American west, the Colorado river provides water to millions of people across the U.S. and Mexico. Its water is fed into agricultural production, industry, and homes. However, as climate change brings rising temperatures, longer droughts, and changing rain patterns, it brings into question the sustainability of the basin's water supply. Simultaneously, our demand for water does not decrease, putting extra strain on the river's resources. Because the U.S. and Mexico are both dependent on the Colorado river, it is important to understand how weather conditions influence water availability and usage across both countries.

The purpose of this project is to investigate the relationship between changing conditions, water availability in the Colorado river, and water use and deliveries between the U.S, and Mexico from 2000 to 2025. The research question is the following: How has decreasing water availability in the Colorado river affected water usage and allocation between the U.S. and Mexico and what role have changing weather conditions played? The reasoning behind starting this study in the 2000 is due to that year being the last year the Colorado river's reservoirs were at full capacity and the river still flowed fully to its terminus in the gulf of California. The central hypothesis is that months with higher temperatures and lower precipitation are associated with reduced Colorado river inflows and lower reservoir levels. Further, periods of lower water availability are associated with changes in U.S. water use and deliveries to Mexico.

The data I will be using to answer the question comes from three different sources. The first is historical records from the U.S. Bureau of Reclamation (USBR) which contains information about Colorado river inflows, reservoir storage, and water use. The second is

weather data from the National Oceanic and Atmospheric Administration (NOAA) which tracks precipitation and drought over the Colorado basin. Lastly, I will be using data from the International Boundary and Water Commission (IWBC) which tracks water discharges and deliveries to Mexico.